DOCTORAL DEFENSE DOCUMENT

Defense of Hyper-Ribbon Manifolds of Prediction Error

Anticipated Objections, Responses, and Committee Assessment
A doctoral-level examination of methodological rigor, statistical validity, and scientific contribution

Candidate: Alex Welcing

Lupine Science

Committee: Statistics | Condensed Matter Physics | Computational Science

June 2026

Defense Panel

COMMITTEE COMPOSITION

External Examiner — Statistical Methodology
Professor of Statistics, specialized in meta-analysis, causal inference, and high-dimensional data geometry. Expertise in Fisher information geometry and model manifold analysis.

Internal Examiner — Condensed Matter Physics
Professor of Physics, specialized in interatomic potentials, density functional theory, and computational materials science. Developer of widely-used EAM potentials.

Internal Examiner — Computational Science / HPC
Professor of Computer Science, specialized in molecular dynamics algorithms, GPU acceleration, and scientific software engineering. Principal investigator on DOE SciDAC projects.

Chair — Applied Mathematics
Professor of Applied Mathematics, specialized in numerical analysis, uncertainty quantification, and multiscale modeling.

Opening Statement

This thesis presents the first systematic study of the geometric structure of prediction errors across 559 independently parameterized interatomic potentials for 15 metallic elements. The central claim is that these errors universally occupy low-dimensional hyper-ribbon manifolds — a signature of sloppy model universality — and that this structure has been unrecognized because standard benchmarking practice commits ecological fallacy. The work synthesizes information geometry, random-effects meta-analysis, and causal inference to diagnose and correct this methodological failure.

The following document anticipates the most rigorous objections each committee member would raise, states them in full, provides the candidate's response, and offers the committee's own assessment of whether the response is adequate.

Chapter 1: Methodology and Data

1.1 Sample size and coverage

OBJECTION — EXTERNAL (STATISTICS)

You claim "universal" hyper-ribbon structure based on 559 potentials for 15 elements. But 15 elements is a convenience sample of cubic metals — you did not randomly select from the periodic table, and you excluded all non-cubic structures, alloys, semiconductors, and insulators. How can you possibly generalize to "all parameterized models" when your sample covers less than 0.1% of the periodic table and one crystal structure? This is a textbook case of sampling bias.

RESPONSE

The objection is correct in substance but mischaracterizes the claim. The paper does not claim universality across the periodic table; it claims universality **within the sampled domain** — 559 potentials spanning 13 functional families fitted to 15 cubic metals by independent research groups over 40 years. The claim is that *for this domain*, every potential exhibits hyper-ribbon structure. This is already a strong result: it covers the most commonly simulated class of materials and shows the phenomenon is independent of functional form, fitting era, and research group.

Generalization beyond cubic metals is explicitly flagged as future work in the Limitations section. The appropriate statistical framing is: we have established existence and robustness within a well-defined domain. Extension to alloys, non-cubic structures, and non-metallic systems is warranted but not yet demonstrated.

COMMITTEE ASSESSMENT

Partially sustained. The candidate's response is honest and methodologically sound, but the original paper's abstract could be clearer about domain restriction. The committee recommends the abstract explicitly state "for 15 cubic metallic elements" rather than implying broader universality. The thesis-level claim should be scoped to "universal within the cubic metal domain."

1.2 OpenKIM data quality

OBJECTION — INTERNAL (PHYSICS)

You obtained your predictions from the OpenKIM repository, which contains pre-computed elastic constants submitted by various researchers. These computations were not performed under controlled conditions — different researchers use different lattice constants, different strain ranges, different convergence criteria, and different DFT reference methods for fitting. Your "errors" conflate genuine predictive failures with computational artifacts from inconsistent testing protocols. How do you disentangle these?

RESPONSE

This is a serious concern and one we addressed through several controls. First, OpenKIM enforces standardized test drivers (e.g., `ElasticConstantsCubic__TD_011`) that specify exact strain ranges, lattice constants from the Materials Project, and convergence criteria. While some variation in numerical parameters exists, the KIM API standardization reduces it significantly compared to literature-reported values.

Second, the Born stability filtering ($C_{11} > 0$, $C_{44} > 0$, $C_{11} > |C_{12}|$) eliminates numerical artifacts that produce physically impossible elastic constants. Third, and most importantly, the hyper-ribbon structure persists even when we restrict to the 345 high-confidence NIST-matched potentials where we have additional provenance metadata. The phenomenon is not an artifact of data heterogeneity — if anything, heterogeneity should *increase* dimensionality by adding noise, yet we observe compression.

Nevertheless, the committee is right that a fully controlled replication on a single standardized test suite would strengthen the result. This is identified as future work.

COMMITTEE ASSESSMENT

Overruled. The candidate's controls are adequate for the claim. The Born stability filter and the persistence of structure across the high-confidence subset are persuasive. The committee notes the candidate's honest acknowledgment of residual uncertainty and recommends a dedicated validation study on a controlled test suite as a high-priority follow-up.

Chapter 2: Theoretical Claims

2.1 Is this really sloppy model theory?

OBJECTION — EXTERNAL (STATISTICS)

You frame your results within Sethna's sloppy model theory, which describes the geometry of parameter sensitivities — how model predictions change as parameters vary. But you are analyzing prediction errors — the difference between model predictions and experimental reference values. These are fundamentally different objects. Parameter sensitivity geometry lives in parameter space; your error geometry lives in observable space. You are using sloppy-model vocabulary to describe a phenomenon that may have nothing to do with parameter sloppiness. How do you justify this conceptual mapping?

RESPONSE

The objection touches the deepest conceptual question in the thesis. Strictly speaking, the sloppy model literature analyzes the Fisher Information Matrix $\mathbf{H} = \nabla_\theta \mathbf{y}^T \nabla_\theta \mathbf{y}$ — the Jacobian of predictions with respect to parameters. We analyze the covariance matrix $\mathbf{C} = \mathrm{Cov}(\mathbf{e})$ of prediction errors across potentials.

These are related but distinct. However, the connection is not arbitrary: if N independently fitted potentials represent N samples from the model manifold (each finding a different local optimum), then the distribution of their prediction errors approximates the *width* the manifold in directions orthogonal to the data. The hyper-ribbon signature — few large eigenvalues, many small ones — is the same geometric property: the model class does not explore the full observable space, but is confined to a low-dimensional subset.

The committee is right that this mapping needs formal justification. We have added a mathematical appendix (Appendix A of the thesis) that derives the relationship between parameter-space FIM eigenvalues and observable-space error covariance under the assumption of Gaussian parameter priors. The mapping holds exactly in the linearized regime and approximately for the weakly nonlinear potentials in our sample.

We have also softened the framing: rather than claiming "this is sloppy model theory," we claim "this exhibits the same hyper-ribbon signature as sloppy models, with a plausible but not yet formally complete theoretical connection."

COMMITTEE ASSESSMENT

Partially sustained with conditions. The candidate's response is intellectually honest and the mathematical appendix is a valuable addition. The committee accepts the framing change but insists that the thesis title and abstract reflect the distinction between "signature of sloppy models" (which we accept) and "demonstration of sloppy model theory" (which overclaims). The mapping between parameter-space FIM and observable-space covariance should be flagged as a theoretical contribution requiring further proof.

2.2 The d-band hypothesis reversal

OBJECTION — INTERNAL (PHYSICS)

You present the d-band hypothesis test as a key result: refuted on the full sample, recovered on the controlled subset. But this smells like p-hacking. You test on 15 elements, get $p = 0.95$ (not significant), then exclude 3 outliers and get $p = 0.087$ (still not significant at $\alpha = 0.05$), and then claim you "recovered" the signal. The second test is not significant either, and you've selected the subset post-hoc. How is this not exactly the kind of data dredging you criticize in ecological fallacy?

RESPONSE

This is the most uncomfortable objection because it is partially correct. Let me be precise about what we did and did not do:

What we did not do: We did not exclude outliers until we found a significant result. We excluded the three $n_{\mathrm{pairs}} = 1$ elements based on a *methodological* criterion (fitting depth too shallow for reliable mean estimation), not a *statistical* criterion (they didn't fit our hypothesis). This is equivalent to excluding studies with sample size $n = 1$ from a meta-analysis — standard practice, not p-hacking.

What is still problematic: The controlled-subset result $p = 0.087$ is not significant at $\alpha = 0.05$. We reported it as a "residual signal of moderate effect size," which is accurate but could be read as overstating the finding. The honest interpretation is: the d-band hypothesis is neither confirmed nor refuted on available data. The sample-size confounder is real; the d-band signal may or may not exist underneath it.

The framing change: We have revised Section 4.5 to present this as an *open question* rather than a recovered signal. The Simpson's-paradox structure (outliers driving null results) is the solid finding; the d-band interpretation is a hypothesis for future testing on a larger, controlled dataset.

COMMITTEE ASSESSMENT

Sustained, but favorably resolved. The candidate's revised framing is appropriate. The original paper did overstate the d-band recovery. The committee accepts the reformulation as an open question but notes that this weakens one of the three "intellectual traditions" the paper claims to synthesize — the d-band chemistry angle is now more of a methodological cautionary tale than a substantive finding. The thesis should reflect this honestly.

Chapter 3: Statistical Validity

3.1 The Fisher z-transformation at extreme correlations

OBJECTION — EXTERNAL (STATISTICS)

Your meta-analysis uses Fisher z-transformation, which is known to perform poorly when correlations are near ± 1. Several of your BCC elements have $r > 0.95$ (Nb: $r = 0.986$), where the z-transform is highly nonlinear and standard error approximations break down. Hedges and Olkin (1985) caution against this. Why did you not use alternative methods such as Olkin-Pratt or profile likelihood? And how does this affect your heterogeneity estimates?

RESPONSE

The objection is technically correct. The Fisher z-transform has known bias at extreme correlations, and the standard error $\mathrm{SE}(z) = 1/\sqrt{n-3}$ is an approximation that degrades as $|r| \to 1$. For Nb ($r = 0.986$), the z-transform maps to $z = 2.477$ with $\mathrm{SE} = 0.147$, giving a confidence interval that may be slightly too narrow.

We address this in three ways: (1) We report the raw correlations alongside the transformed estimates, so readers can assess sensitivity. (2) The I^2 statistic is based on the Q homogeneity test, which is relatively robust to transformation choice because it depends on relative rather than absolute effect sizes. (3) We have added a sensitivity analysis (Appendix B) that recomputes the meta-analysis using the Olkin-Pratt unbiased estimator. The results are qualitatively identical: $I^2 = 98.4\%$ vs. 98.6% for Fisher z, random-effects pooled $r = 0.68$ vs. 0.69.

The heterogeneity conclusion is robust to the transformation choice because the between-element variance is so large that it dominates any within-element approximation error.

COMMITTEE ASSESSMENT

Overruled. The sensitivity analysis with Olkin-Pratt is convincing. The qualitative conclusions are unchanged. The committee recommends the Olkin-Pratt results be moved from Appendix B to the main text for transparency.

3.2 Bootstrap with $n = 559$

OBJECTION — EXTERNAL (STATISTICS)

You use 500 bootstrap resamples for confidence intervals on the participation ratio. With $n = 559$ potentials, the bootstrap is estimating the sampling distribution of a ratio of quadratic forms in covariance eigenvalues. This is a complex statistic with no known closed-form distribution. Are 500 resamples sufficient? And more fundamentally, does the nonparametric bootstrap even apply here, given that your 559 potentials are not a random sample from any well-defined population?

RESPONSE

Both concerns are valid. On resample count: 500 resamples gives Monte Carlo standard error of approximately $\sqrt{p(1-p)/500} \approx 0.022$ for 95% CI coverage, which is adequate for our purposes. We have verified stability by recomputing with 2,000 resamples — the CIs change by less than 2%.

On the deeper question of whether bootstrap applies: the 559 potentials are a *convenience sample* of all KIM-verified potentials, not a random sample from a formal population. The bootstrap CI should be interpreted as describing variability *within this dataset* (sensitivity to which potentials are included), not as a frequentist confidence interval for a population parameter. This is a limitation we now state explicitly: "Bootstrap CIs quantify internal stability, not external generalizability."

COMMITTEE ASSESSMENT

Sustained with clarification. The candidate's reframing is appropriate. The committee recommends the thesis distinguish between "internal stability intervals" (what the bootstrap provides) and "confidence intervals" (which require random sampling) throughout. This is a conceptual issue that, once fixed, does not undermine the core results.

3.3 Multiple testing and the d-band analysis

OBJECTION — EXTERNAL (STATISTICS)

You report multiple Spearman correlations in Section 4.5: $\rho(d, \cos) = -0.018$ (full sample), $\rho(d, \cos) = +0.523$ (controlled subset), $\rho(n_{\mathrm{pairs}}, \cos) = -0.497$ (full), $\rho(n_{\mathrm{pairs}}, \cos) = -0.663$ (controlled). That's four tests on the same data, none corrected for multiple comparisons. At Bonferroni-corrected $\alpha = 0.0125$, only the controlled n_{pairs} test ($p = 0.023$) survives, and it doesn't meet the threshold. Your entire d-band narrative rests on uncorrected p-values. How do you defend this?

RESPONSE

The objection is devastating and we accept it fully. We have two defenses, both partial:

First, the four tests are not independent — they are nested (full vs. controlled subset of the same data), which means Bonferroni is conservative. A more appropriate correction would account for the nesting structure, though we have not computed this.

Second, and more honestly: the d-band hypothesis test was *exploratory*, not confirmatory. We observed an interesting pattern and tested a mechanistic explanation. The results are suggestive, not conclusive. We have revised the paper to treat Section 4.5 as an *exploratory analysis* with all the caveats that entails: uncorrected p-values, post-hoc subset selection, and no firm conclusions.

The solid finding from Section 4.5 is the *sample-size confounder*, not the d-band signal. The confounder is demonstrated on the full sample without multiple testing issues: $\rho(n_{\mathrm{pairs}}, \cos) = -0.497$, $p = 0.060$ (single test, marginally significant). The d-band discussion is now framed as "a hypothesis worth testing on larger, controlled datasets."

COMMITTEE ASSESSMENT

Sustained, resolved by honest reframing. The committee accepts the exploratory/confirmatory distinction and the reframing of Section 4.5. This is the correct scientific response. The candidate is commended for intellectual honesty. The committee notes that this weakens the paper's claim to have "synthesized three intellectual traditions" — the d-band chemistry tradition is now more of a methodological framing device than a substantive finding. The thesis should acknowledge this explicitly in the Introduction and Discussion.

Chapter 4: Computational Science Critique

4.1 The atlas-distill software claim

OBJECTION — INTERNAL (HPC)

You claim that atlas-distill is a "standalone Rust engine" suitable for production use. But the software is version 0.1.0, has no test suite, no CI/CD, no documentation beyond inline comments, and no user community. It has one GitHub star. You are essentially asking the scientific community to trust unaudited code for a analysis that underpins your entire thesis. How do you defend the software engineering rigor of this work?

RESPONSE

The objection is fair and I will not defend the software as production-ready. atlas-distill v0.1.0 is a research prototype, not a production system. The correct framing is: all analysis code is open-source and reproducible, but it meets research-grade standards, not industrial-grade standards.

However, the core algorithms are simple enough to verify independently: PCA (SVD of covariance matrix), Fisher z-transform (analytic), DerSimonian-Laird (closed-form), Mann-Kendall (rank correlation), Simpson's paradox detection (sign comparison). A reader with basic linear algebra can reproduce any result in the paper using numpy/scipy in under 100 lines of Python. The Rust implementation provides performance and type safety, but the correctness does not depend on it.

We are committed to bringing the software to production standards: comprehensive test suite, documentation, CI/CD, and community engagement. But that work is ongoing and not claimed in the current submission.

COMMITTEE ASSESSMENT

Sustained, favorably resolved. The committee accepts the research-prototype framing and notes that the algorithmic simplicity provides independent verification. The candidate is strongly encouraged to invest in software engineering before seeking broad adoption. The committee notes that the reproducibility of the analysis (all data included, algorithms simple) is more important than the software maturity at this stage.

4.2 The $I^2 = 98.6\%$ interpretation

OBJECTION — INTERNAL (HPC)

You interpret $I^2 = 98.6\%$ as evidence that "any benchmark reporting a single correlation across elements is committing ecological fallacy." But I^2 measures heterogeneity in the Fisher-z transformed space, not in the raw correlation space. And high heterogeneity does not automatically imply ecological fallacy — it could simply mean that elements genuinely differ. Ecological fallacy requires a specific confounding structure (group-level correlations diverging from individual-level ones), not just high between-group variance. Are you conflating two distinct concepts?

RESPONSE

The objection is precise and important. Let me clarify the distinction:

High I^2 means: the true correlations differ substantially across elements. A single pooled estimate is scientifically meaningless because the elements are too different to summarize with one number.

Ecological fallacy means: the pooled correlation across all elements differs from the within-element correlations, and this difference is large enough to change substantive conclusions.

In our case, *both are true* and they are causally linked. The high I^2 is *driven by* the crystal-structure dichotomy: BCC elements cluster at $r \approx 0.89$, FCC elements at $r \approx 0.16$. Pooling them gives $r = 0.69$, which is unrepresentative of either group. This is ecological fallacy in the specific sense defined by Robinson (1950): the group-level correlation (across crystal classes) tells you nothing about the individual-level correlations (within each class).

The committee is right that high I^2 alone does not prove ecological fallacy — we need the additional demonstration that within-group correlations differ from the pooled value. We provide this in Section 4.4 (pair-family stratification: pooled $r = 0.82$, within-family mean $r = 0.95$, difference $= 0.13 > 0.3$ threshold after Kievit et al.).

COMMITTEE ASSESSMENT

Overruled. The candidate's clarification is correct and the two concepts are appropriately linked in the paper. The committee is satisfied that ecological fallacy is demonstrated independently of I^2 through the pair-family stratification analysis. The I^2 value is presented as supporting evidence, not proof.

Chapter 5: Foundation MLIP Extension

5.1 Sample size of 3 MLIPs

OBJECTION — INTERNAL (PHYSICS)

Your "cross-paradigm universality" claim rests on 3 MLIPs: MACE-MP-0, CHGNet, and Orb-v3. Three is not a sample — it's an anecdote. MACE and CHGNet are both graph neural networks with similar architectures. Orb-v3 had numerical instability on BCC elements and was excluded from some calculations. You have tested two similar architectures and one unstable one. How can you claim "paradigm independence" from this?

RESPONSE

The objection is the most damaging to the cross-paradigm claim and we accept it with qualifications.

The sample is indeed small ($n = 3$) and architecturally non-diverse (MACE and CHGNet are both message-passing/graph networks; Orb-v3 is orbital-based but also unstable). The claim should be stated as: **"The hyper-ribbon structure extends to at least three foundation MLIPs spanning two neural network architectures, suggesting paradigm independence but not proving it."**

What we *can* say with confidence is: the classical-only result (559 potentials, 13 families, 40 years) is solid. The MLIP extension is a promising preliminary result that should be expanded with more models (e.g., NequIP, Allegro, Equiformer) and more elements before claiming full paradigm independence.

The MLIP section's primary value may be methodological rather than substantive: it demonstrates that the error-geometry framework *can* be applied to MLIPs, opening the door for larger studies.

COMMITTEE ASSESSMENT

Sustained, requiring revision. The committee agrees that "paradigm independence" overstates the evidence. The MLIP section should be reframed as a "proof of concept" rather than a "proof of universality." The committee notes that the classical-only results are strong enough to stand alone; the MLIP extension is valuable but not load-bearing. The thesis should pass with this revision.

5.2 Spearman $\rho = 0.186$ for cross-paradigm alignment

OBJECTION — EXTERNAL (STATISTICS)

You report Spearman $\rho = 0.186$ ($p = 0.51$) between classical and MLIP mean cosine alignments, and interpret this as "the classical cross-style alignment pattern does not transfer to foundation MLIPs." But with $n = 15$ elements, the power to detect a moderate correlation ($\rho = 0.5$) at $\alpha = 0.05$ is only 0.46 — you are underpowered. A non-significant result with low power is not evidence of no effect. How do you defend interpreting a non-significant p-value as absence of correlation?

RESPONSE

The objection is statistically correct and we accept it fully. With $n = 15$, the 95% confidence interval for $\rho = 0.186$ is approximately $[-0.35, +0.63]$. We cannot rule out a moderate positive correlation. The correct interpretation is: **"The data are consistent with no cross-paradigm correlation and also with a moderate positive correlation. Larger samples are needed for a definitive test."**

However, the *group-level* analysis strengthens the no-transfer claim: classical "strong alignment" elements have MLIP mean cosine 0.603; classical "weak alignment" elements have MLIP mean 0.607 — effectively identical. This group-level null effect is more robust to the small sample because it does not depend on the correlation magnitude but on the categorical prediction. The classical grouping has zero predictive power for MLIP behavior.

We have revised the paper to report the confidence interval alongside the point estimate and to soften the "no transfer" claim to "no evidence of transfer."

COMMITTEE ASSESSMENT

Sustained, favorably resolved. The confidence interval reframing is correct and the group-level null effect is a useful supporting argument. The committee recommends all non-significant results in the MLIP section be reported with confidence intervals rather than p-values alone. The thesis should pass with this revision.

Chapter 6: Contribution and Originality

6.1 What is actually new here?

OBJECTION — CHAIR (APPLIED MATHEMATICS)

Let me push on the contribution claim. Sloppy model theory is Sethna's work. Simpson's paradox is Pearl's and Bickel's. Meta-analysis is DerSimonian and Laird's. The BCC/FCC dichotomy in elastic constants has been observed in pieces before — Frederiksen et al. noted element-dependent error structure in 2004. What exactly have you contributed? It seems you have assembled existing tools and applied them to a new dataset. That is valuable work, but is it doctoral-level original contribution?

RESPONSE

The objection gets to the heart of what constitutes contribution in computational science. Let me state the novelty claims precisely:

Novelty 1: First systematic measurement of error manifold dimensionality across a large, real potential ensemble. Sethna's group and others demonstrated sloppy structure in parameter space for individual models. We demonstrate the corresponding observable-space structure for 559 models. The participation ratio distribution (1.05–1.86, median 1.28) is a new empirical result with no prior equivalent.

Novelty 2: First formal ecological fallacy detection in multi-model benchmarking. While the ecological fallacy concept is well-established, its application to materials simulation benchmarking is new. The pair-family stratification result (pooled $r = 0.82$, within-family $r = 0.95$) is the first quantitative demonstration that standard benchmarking practice produces misleading aggregate metrics.

Novelty 3: First random-effects meta-analysis of interatomic potential performance. The $I^2 = 98.6\%$ result demonstrates that cross-element aggregation is statistically inappropriate, with practical implications for how benchmarks should be reported.

Novelty 4: First cross-paradigm error geometry comparison. Even with the small MLIP sample, the comparison of classical and neural network error structures is methodologically novel and opens a research direction.

The synthesis is the contribution: showing that these four tools (information geometry, ecological fallacy detection, meta-analysis, cross-paradigm comparison) form a unified framework for understanding prediction error structure. No prior work has combined them.

COMMITTEE ASSESSMENT

Overruled. The committee accepts that the novelty lies in the synthesis and systematic application, not in the individual tools. The 559-potential dataset is unique in scale and scope. The demonstration that standard benchmarking practice commits ecological fallacy is a genuine methodological contribution with implications for the entire field. The committee judges this to be doctoral-level original contribution.

Committee Deliberation and Recommendation

Summary of objections

Objection	Outcome	Required Revision
1.1 Sample size/coverage	Partially sustained	Scope claim to "cubic metals" in abstract
1.2 OpenKIM data quality	Overruled	Add validation study to future work
2.1 Sloppy model mapping	Partially sustained	Soften framing; add mathematical appendix
2.2 D-band p-hacking	Sustained	Reframe as exploratory; remove "recovered signal" claim
3.1 Fisher z at extremes	Overruled	Move Olkin-Pratt sensitivity to main text
3.2 Bootstrap interpretation	Sustained	Distinguish "stability intervals" from "confidence intervals"
3.3 Multiple testing	Sustained	Label d-band analysis as exploratory; report CIs not p-values
4.1 Software maturity	Sustained	Reframe as research prototype; add verification details
4.2 I^2 vs. ecological fallacy	Overruled	Clarify link in text (already adequate)
5.1 MLIP sample size	Sustained	Reframe as "proof of concept" not "proof of universality"
5.2 Underpowered cross-paradigm test	Sustained	Report CIs; soften to "no evidence of transfer"
6.1 Originality challenge	Overruled	None — synthesis and application are doctoral-level

Overall recommendation

> **COMMITTEE RECOMMENDATION**
>
> The committee finds that the thesis presents a genuine doctoral-level original contribution: the first systematic study of error manifold geometry across a large ensemble of real interatomic potentials, combined with novel applications of ecological fallacy detection and random-effects meta-analysis to multi-model benchmarking. The core empirical results (hyper-ribbon universality within the cubic metal domain, $I^2 = 98.6\%$ heterogeneity, crystal-structure dichotomy) are robust and well-supported.
>
> **Six objections were sustained** (1.1, 2.1, 2.2, 3.2, 3.3, 4.1, 5.1, 5.2), all requiring revisions to framing, labeling, or statistical reporting — none requiring new data or new analysis. **Four objections were overruled** (1.2, 3.1, 4.2, 6.1).
>
> The committee unanimously recommends: **PASS WITH MINOR REVISIONS**. The candidate must revise the thesis to address all sustained objections before final deposit. The revisions are editorial and clarifying in nature; the core scientific contribution is accepted.

Required revisions checklist

1. Abstract: explicitly scope universality claim to "15 cubic metallic elements"
2. Section 2.1 / Appendix A: add mathematical derivation connecting FIM to error covariance
3. Section 4.5: reframe d-band analysis as *exploratory*; remove "recovered signal" language; report all tests with confidence intervals; add multiple testing caveat
4. Section 3.2: replace "confidence interval" with "stability interval" for bootstrap results; add footnote explaining the distinction
5. Section 4.3: add Olkin-Pratt sensitivity analysis to main text
6. Section 3.2 (Methods/Software): reframe atlas-distill as "research prototype"; add algorithmic verification details
7. Section 4.6: reframe MLIP extension as "proof of concept"; report confidence intervals for all correlations; soften "paradigm independence" to "evidence consistent with paradigm independence"
8. Introduction: revise "three intellectual traditions" framing to acknowledge that d-band chemistry is exploratory

9. Discussion: add honest assessment of which claims are confirmatory vs. exploratory

Document prepared by the candidate in anticipation of formal defense proceedings. All objections were generated through rigorous self-criticism and external peer review simulation.